

CHEM 101 Topic 10: Chemical Bonds — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. For electron dot structures, ensure all valence electrons are accounted for and octets are satisfied. For geometry problems, identify the number of bonding and lone pairs before naming the shape.

Section 1: Ionic and Covalent Bonds

Problem 1: Determine if the bond in K_2O is ionic or covalent.

Problem 2: Compare the expected melting point and solubility of NH_3 (covalent) versus NaF (ionic).

Problem 3: Explain the difference between a "molecule" and a "crystal lattice." Which one is formed by ionic bonds?

Section 2: Electron Dot Structures

Problem 4: How many total valence electrons are available for the molecule PCl_3 ?

Problem 5: Draw the electron dot structure for H_2S . (S is the central atom).

Problem 6: Draw the electron dot structure for HCN (Hydrogen Cyanide). Be sure to include any necessary multiple bonds to satisfy octets.

Section 3: Molecular Geometry (VSEPR)

Problem 7: What is the molecular geometry of a molecule with 2 bonding pairs and 0 lone pairs on the central atom (e.g., CO_2)?

Problem 8: Predict the shape of NH_3 . (Hint: N has 3 bonding pairs and 1 lone pair).

Problem 9: Predict the molecular geometry of CH_4 .

Section 4: Coordinate Bonds & Polarity

Problem 10: What is a coordinate covalent bond, and how does it differ from a standard covalent bond in terms of electron contribution?

Problem 11: Compare the electronegativities of Nitrogen and Hydrogen. Is the N-H bond polar or non-polar?

Problem 12: Water (H_2O) has polar O-H bonds and a "Bent" geometry. Does this make the overall water molecule polar or non-polar? Explain why.